

# Python Programming Assignment

Real-World Scenarios: Variables, Input/Output & Operators

Course: Python Fundamentals

Total Scenarios: 20

Focus: Core Operators & Basic Logic

## 1. Shopping Bill

Operators: +, -

A customer buys a shirt for ₹800 and shoes for ₹1,200. The shop gives a discount of ₹150.

**Task:** Write a Python program to calculate and display the final bill amount.

## 2. Movie Ticket Calculation

Operator: \*

A movie ticket costs ₹180. A family has 5 members.

**Task:** Write a Python program to calculate the total ticket cost.

## 3. Employee Salary

Operators: +, -

An employee's monthly salary is ₹25,000. They receive a bonus of ₹3,500 and have a deduction of ₹1,200.

**Task:** Write a Python program to calculate the final salary.

## 4. Restaurant Bill Sharing

Operator: /

Four friends go to a restaurant. The total bill is ₹2,400.

**Task:** Write a Python program to calculate how much each person should pay.

## 5. Pizza Sharing

Operators: //, %

A pizza has 25 slices. Five friends want to share the slices equally.

**Task:** Write a Python program to calculate how many slices each friend gets and how many slices are left.

## 6. Convert Minutes to Hours

Operators: //, %

A movie lasts 185 minutes.

**Task:** Write a Python program to convert the duration into complete hours and remaining minutes.

## 7. Student Marks Calculation

Operators: +, /

A student scored English = 85, Maths = 90, Science = 80.

**Task:** Write a Python program to calculate the total marks and average marks.

## 8. Electricity Bill

Operator: \*

A household consumed 250 units of electricity. Each unit costs ₹6.

**Task:** Write a Python program to calculate the total electricity bill.

## 9. Age Calculation

Operator: -

A person was born in 2005. The current year is 2026.

**Task:** Write a Python program to calculate their age.

## 10. Simple Interest

Operators: \*, /

A person deposits ₹50,000 in a bank. Principal = 50000, Rate = 6, Time = 2.

**Task:** Write a Python program to calculate Simple Interest using  $SI = (P * R * T) / 100$ .

## 11. ATM Withdrawal

Operator: -

An ATM account has ₹10,000. A customer withdraws ₹3,750.

**Task:** Write a Python program to calculate the remaining balance.

## 12. ATM Note Calculation

Operators: //, %

A customer wants to withdraw ₹4,750.

**Task:** Write a Python program to determine the number of ₹500 notes and the remaining amount after ₹500 notes.

## 13. Mobile Recharge

Operator: -

A customer has ₹500 in their wallet. They purchase a mobile recharge for ₹249 and a data pack for ₹99.

**Task:** Write a Python program to calculate the remaining wallet balance.

#### 14. Discount Calculator

Operators: \*, /, -

A laptop costs ₹60,000. The shop provides a 10% discount.

**Task:** Write a Python program to calculate the discount amount and final price.

#### 15. Travel Distance

Operator: \*

A car travels at a speed of 60 km/h for 4 hours.

**Task:** Write a Python program to calculate the total distance travelled using  $\text{Distance} = \text{Speed} \times \text{Time}$ .

#### 16. Temperature Conversion

Operators: \*, /, +

Write a Python program to convert Celsius into Fahrenheit using  $\text{Fahrenheit} = (\text{Celsius} * 9 / 5) + 32$ .

**Example:** Celsius = 30.

#### 17. Even or Odd

Operator: %

A school wants to check whether a student's roll number is even or odd.

**Task:** Write a Python program that takes a roll number and checks whether it is even or odd using the modulus operator.

#### 18. Compare Product Prices

Operators: <, >, ==

Two online stores sell the same mobile phone. Amazon Price = 24999 and Flipkart Price = 23999.

**Task:** Write a Python program using comparison operators to determine whether Amazon is cheaper, Flipkart is cheaper, or both have the same price.

#### 19. Student Eligibility

Operators: >=, and

A college allows a student to participate in an exam if attendance is greater than or equal to 75% AND marks are greater than or equal to 40.

**Task & Example:** Write a Python program to check whether the student is eligible. (Example: Attendance = 82, Marks = 55).

## 20. Online Shopping Offer

Operators: `>=`, `and`

An online shopping website provides a discount when the order amount is ₹2,000 or more AND the customer is a premium member.

**Task & Example:** Write a Python program to check whether the customer gets the discount. (Example: Order Amount = 2500, Premium Member = True).